ActInf Textbook Group ~ Cohort 4 ~ Meeting 7 (Chapter 3 part 1)

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hello it's
July 18th and we're in our first
discussion of chapter three
in cohort 4. so before we go to any
questions or anything does anyone want
to just
bring up any overall thoughts they had
on chapter three
just about the chapter where it is in
the book
or what they
remember or got from that chapter
yeah I would add that I actually just
generally think the chapter was written
very well and it was very clear
um I think the high road can be
philosophically and mathematically very
abstract
so I personally appreciated the kind of
clarity that the authors took to
discussing things like Marco blankets
and um
existential imperatives I guess
it's it's a the high road
perhaps more than low road provokes
further questions further ontological
um and philosophical metaphysical
questions
you know
the well what you know how how deep does
the hierarchy of Mark of blankets run or
what's the fundamental base layer of
this
um this whole system
but on the whole at least in so far as

the free energy principle is a very
simple explanation of something very
fundamental
um yeah I thought they did a really good
job outlining that particular point
nice thank you anyone else any thoughts
low road High Road anything yeah yeah I
really enjoyed um this chapter probably
because there's less equations to uh get
hung up on but especially section 3.7
just the idea of using active inference
to reconcile
um different theoretical perspectives so
in my area action we have debates about
whether
movement commands are represented
centrally versus self-organizing or are
inactive this is hotly debated for for
decades now
um that that's something that's really
attracted me to active inference is to
have more unifying view
um
rather than keep going with the debating
so I really appreciated section 3.7 for
me personally
any other thoughts on three or about how
somebody is seeing the low road and the
high road
well Daniel I want to see um about the
the figures when I when I look at the
figures
uh for example a figure 3.1
where uh
uh where the figure explains the uh
blanky stays
um but in addition to this Blank Space
there are um many other

like notations and and formulas
um
so it seems quite quite uh intimidating
those foreigners
like like uh
for example the the first equation on
the left you have X you know dot x equal
to f of x Etc so others all this
notation is not explained that's you
know it seems to me
um that it creates uh blocks for
understanding
um of the uh Concepts
um
so so Vision I thought it's it's kind of
a little bit incompanied uh I know it's
difficult because those are those in
order to understand these Concepts you
need more uh
background uh explanations
um
so so but when I read those figures I I
hope to understand
all the information that it contains in
the figures if there are things that not
understandable
uh that it creates a kind of a mental
block uh in my mind
sure
totally makes sense so and it's good to
want to understand all the notation
so this is a representation that is
going to come up again and again
um it's the Markov blanket and there's
kind of two
uh
types of notation happening here there's
like the variables that are being

locally assigned
and then there's some more conventional
notation
so like some of the conventional
notation is f is a function of something
and then
Omega is a noise term on something
that's conventional and also the dot
being uh rate of change so it's kind of
like having like a prime or a first
derivative and then there's the locally
assigned variables so external X
internal mu
octave U sensory y
and then
um now there is a typo in this figure
um
a small one
just
it's in the erata
okay
but
um
the visuals are correct
each of these functions
each the rates of change of each of
these four kinds of states
we're describing the rate of change
so it's kind of like a flow map for all
four of these nodes on a graph
for the internal external states in the
blanket right rate of change yeah and
each of those rates of change is broken
up into two components but kind of like
a signal and a noise
that's and noise could be zero or it
could be all noise
and then signal components on the rate

of change of each of the
[Music]
um
nodes in this particular partition
the signal component is a function of
all the incoming arrows
and itself at that time
right
so so in chapter two we we understand
there are no road we understand the
active inference as the kind of
optimization problem where you can you
have a choice of action and and
perception so how how does actually in
the perception
play a role in this figure in the in the
uh
in this Mark of blanket in this uh in a
figure you just showed
yeah good
question well we associate
this incoming sensory state
according uh to perception
and then the outgoing statistical
dependencies
according to action
and then what happens here which could
be nothing if it's a very simple system
or it could be sophisticated or unknown
the same as external
um we just call this the cognitive
States so this is the perception
cognition and action
of the agents
blanket plus internal States that's the
particular thing
uh uh insulated off from the external
States

so that's why this is like the minimal
base graph
for the cybernetic agents
is having the input cognition and output
and the um the work of isomera and first
and others like Maps this onto neural
networks that have like an input
processing and output layer
right okay
everything what are the choices because
in the chapter two it talks about the
the perception and the action are the
choices made by the age out right right
so those are the choices not the states
and it's a decision kind of decision you
can make you can choose to to to to to
precept this way or the other way you
can choose to take that action or or
another action
so here
you're you're seeing the action state is
is the action is the choice
to um ways to think about this first is
that's why we talk about beliefs about
action action prior uh policy prior
policy posterior so because we're
talking about our beliefs about action
and the second thing is
this is coming from a high roads this is
kind of like a
forget what you knew about chapter one
and two start with just this particular
partition on a base graph
and then the chapter continues
with connecting it more to perception
action
but start with like this is the topology
of systems that can um sense and act

or you could propose a different
topology of that
system
um young and then anyone else
uh yeah uh I just wondered so the
sensory states are the perception and
the active states are the action
intuitively and well those two are
directly connected to each other
and which is which means that they can
bypass the cognition and it's really
hard for me to imagine a state of at
active States just bypassing the
Coalition and just going directly to the
sensory State and also vice versa so can
anyone come up with a good example of it
awesome question does anyone want to
give a thought on that about this what
is this double-headed arrow in the
center
um
so my understanding this error means
that the active States is affected by
by the uh sensory data at your connect
so basically you can see here fu is a
function of Y why is the uh sensory data
that the agent gathered from
um from the external world together for
me yeah
and on the other hand also the U is the
your action
also will affect your uh sensory data y
uh for example in the in the chapter two
the apple and frog example
if you choose your action to look at
apple tree
that's you if you move your eye towards
Apple chain that will generate

scissoring data about Apple you know
what it Observer will be apple
so that would be the action will affect
your sensory data why
and you essentially did a while I will
affect your action
uh yeah I'm just thinking
um
these are all
true
um another thing to keep in mind is like
the effect of a given
Edge in a model or system of Interest it
might not be relevant but it's kind of
like in principle so if you have a
computer system where one computer is
getting input and sending it to a second
one sending it to a third one and then
emitting it there is no effect of this
Arrow or if you have some you know some
other system where this in practice is
zero so just because there's an edge
doesn't mean that in in that simulation
or that model it's like it matters in a
certain way
and the reason why it's there in
principle is because the definition of
the markup blanket
for a given node
is all the nodes that kind of insulate
it and the co-parents
and so just in principle they could be
connected
under the markup blanket uh definition
like the Marco blanket nodes could all
be connected to each other and be
connected into the
um node of Interest

and also these assignments of internal
external active sensory
these are all relative to a given
internal node of Interest
from if you call this one internal
then this would be sensory and that
would be active
or if you picked one over here it would
be different so these are not like
essential features of nodes these are
just with respect to a chosen
internal State on a created model
um Darius
yeah I mean um
just staying on the market blanket for a
bit
um I've seen quite a lot of conflicting
literature and opinions on to what
degree it constitutes a metaphysical
reality
so I think one can't help but think of a
kind of cell when you think of a Markov
blanket and the blanket states that are
the boundary between the cell and the
external environment
to a degree
or Mark or blankets from the biggest
markup blanket
a universe with the universe all the way
down to the smallest of Markov blankets
and all the ones embedded in between
to what degree do current
um free energy principle Ferris believe
that there's a genuine metaphysical
reality or is it just a way of
visualizing and understanding
statistical regularities
yeah big question what does anyone think

budget or anyone else
um I guess like I would think it would
uh
uh so would that
be connected to like our subjective
experience would be
this statistical model so that would be
the Markov blanket
uh
is that what you mean or I mean
I don't mean that it's Observer uh
relative
you have the universe as is
and our observations indicate to us that
there are
um separate
patterns of existence things that are
distinct from one another
uh-huh all those boundaries which are
defined by the markup blanket
metaphysical as if they exist
I mean I think the only way of kind of
conceptualizing that is as a material
distinction or do they just exist do
they solely serve
to be a statistical representation of
the actual distinctions that we observe
but perhaps there is no physical
substrate underlying those regularities
that's my question
yeah it's a huge question there's
there's a ton to say in it definitely a
lot of perspective so like there's a way
broader um Range
one topic that comes up in that setting
is the different flavors of
instrumentalism and realism and
scientific realism pragmatism

so instrumentalism is like the least
metaphysically committed
and just says look linear regressions
help us understand the relationship
between this and that
that gives a sort of cognitive realism
to the linear regression it might
influence someone's life so it does
matter so it doesn't mean that we don't
take it seriously but
no one is going to mistake the map for
the territory
um then for various uh reasons
there's uh also more of a realist
not necessarily like a platonic realist
but like a scientific realism it's like
yeah the objects that are under study
are real
now the finger isn't the moon so
pointing at the phenomena it's still
like not map territory the model and the
equations
is not a system
that's obvious so just saying but the
model doesn't have this feature of the
system is like yes but we're talking
about the reality of this system you
know six people in the elephant they can
still be talking about the reality of
the elephants that's kind of a
scientific realism
um
and uh Mel andrews's work and the math
is not the territory live stream 14 and
a lot of other discussions on that and
then
um there's a lot more to say but that's
a great question area

friend
um
I was when I read this chapter I I met
uh the example that I thought oh well I
intuitively kind of found helpful was
um
playing the piano
and I wondered if you could verify
uh the veracity of that example if it
doesn't indeed explain many of the
Dynamics between my internal States my
active States the hidden States and in
turn the sensory states that I received
to complete an action uh kind of
feedback loop and I think it also might
explain the uh
um The Edge is between
um the active States uh in the example
of playing the piano and uh the sensory
state that might ensue should you take a
particular action all in this instance
as I hope to explain not take a
particular action so I I thought of the
piano kind of sitting at the piano with
the intention of playing and I
the the I believe I want to play piano
I'm not 100 certain of what might be
fall under you know what can um
would
fall under the internal States but other
than a belief that I can play I'm sure
there are many many other factors there
that would be
um necessary for one to
um play initiate the action but then I
touch a key a a note and that is that
has
that entrains an effect on the

perception of the piano which then in
turn I don't I don't play that note
directly
um I elicit sound from that from the
piano which then in turn uh gives me the
noise which hopefully sounds good
and so I
um continue that
that effect by playing the next notes in
accordance with the previous one and so
I build the tune as I go hopefully
and hopefully that's pleasing rather
than full of kind of dissonance and
offbeats and and irregularities and I
thought about the relationship between
the active States and the well and the
and the the blanket States the sensory
States if you were to Omit a note
they buy inhibiting the effect
of the action on the piano thereby if
you won't perceive anything in its you
won't get anything in return so to speak
from the piano because you have emitted
a note
um
so yeah I kind of
um
thought about this around playing the
piano
and
yeah
awesome
that's a really cool example and um in
chapter seven
there's gonna be like a musical example
and music cognition there's a lot more
um
work on as well but in this example like

the wrong note is played
quote wrong note different than the
expected tune but the expected the
expected note is still heard because the
prior is so strong and so like um Ali
and and several others have have a lot
of interest in in that um
music perception it's like very rich
with with music perception and
generation improvisation
it's a really cool area
Daniel if I can just ask if you can go
back to diagram 3.1 I think it was I
just wanted to clarify something that's
just come to mind now and I'm not very
good at all the uh all the technical
details yet but how is a uh the noise
from the external States
how does that
uh
how does that relate to how does a
correlate correlate to the impact or the
effect of the noise at the sensory
states is it a
how is that kind of transduced I'm
thinking of the piano example again
um how does that dissipate through the
system
is it equivalent so a particular
um noise at the external state is it how
is that experienced
or understood as
how is it similar or dissimilar from the
the noise perceived at the
um or generated in the sensory state
yeah
very interesting question
so think about each of these four nodes

as being like a sensor or a thermometer
um in in and of themselves like their
probes that are tracking these flows so
what would it mean to have
um this be high and this below well
maybe the um temperature in the
ecosystem is constant but it has a lot
of variability so there's like no signal
really but there's a high variability
but we have a very accurate thermometer
that we're taking in information from or
conversely maybe it's a deterministic
temperature outside
and our thermometer is highly variable
so this is kind of just like the
outline
that gets filled in like a hundred
percent during the modeling process
but this is just extremely General
outlining like where the city just
saying that each of them are partitioned
into a signal and noise component in
principle
but but that's why it really helps to
also to think about specific examples
and um
yeah
um uh or go ahead oh yeah
um like because I'm a psychiatrist uh uh
what what keeps in my to my mind about
this micro blanket is that it's really a
cool concept you think of the process of
psychoanalysis because
like the therapists by using by using
his own internal State and inferring the
internal state of the patients added by
making decisions and influencing these
sensory States and these active States

active States active States active
States activities
to change the internal State or the
model internal model of the patient so I
mean that's quite interesting because
like the basic concept of Marco blanket
is that you can't really have access to
external States
which means that you don't have any
access to the internal state of others
but you can change the internal state by
having very good inference about it and
doing the right active actions or
providing good sensory States like make
maybe making the patient comfortable or
sometimes you you use
um many different techniques with
Psychotherapy such as hypnotics or
um
I don't know maybe other relaxation
techniques so well
the basic concept Market blanket is
really cool for me so like it wasn't a
question it was just Electronics
that's awesome thank you
[Music]
um
that is absolutely the core of the
Markov blanket concept definitely a lot
of philosophical like
smoke and fire is in that area for for I
think a variety of interesting reasons
partially because it touches on like how
things really are in some people's eyes
um
or it's something that seems to explain
like everything and nothing but at the
end of the day

it's just a description of the sensorium
and the real important
um missing the the missing piece or the
missing Edge not missing piece that
really embodies the constraints here is
is the no telepathy no telekinesis
clause
no Edge between internal and external
States except via
insulating nodes
so by making a fully insulating
shell
then you you don't need to make any
claims about what's in the on the other
side of the blanket
now you could you might happen to know
or you might have a really good model of
what is happening or something like that
but those models aren't free information
takes energy
um and they're always understood as
projections of of onto hidden States or
trackings of hidden States
so it's really just a recognition of the
sensorium and the cybernetic system
boundaries with actually a lot of
generalizations that make it really
exciting and even more flexible than
people might initially
consider just from looking at this
relatively simple system Darius
yeah I was um I was wondering this seems
for this makes perfect sense to me in
terms of a sentient uh non-steady State
equilibrium system like a human brain
um but if in terms of this equation in
terms of the markup blanket generally
but to what degree can we think that

entities that don't act that exist in
steady state equilibria so you know like
for instance classic example of a drop
of oil in water
can we I understand that you can't
necessarily attribute action to the drop
of oil
um or at least purposeful action when we
look at that diagram in 3.1 do we have
to think of it as if the drop of oil
it's as if they're behaving in this
manner
or are we committed to some notion of
course the external and the internal
state
but they're maximizing its model
evidence it's actually actually is
sensing and it is being active so where
where is the kind of
dividing line here between sort of
sentience and and non-sentience and to
what degree if there isn't if there is
one is it just that the drop of oil
basically exists it just seems to be
maximizing its model evidence
but there's no real tell us to it
um
yeah you know that because the action
here the active inference seems to me to
be very highly linked to sentience
yeah
great great questions well one angle is
an instrumentalist goes into that area
without feeling like they're taking on
um uh descriptions like it's a sentient
as they extract the signal of so that's
very like relational or sort of like
perspectival view on sentience

um where somebody taking on more
ontological commitments would say that
they are
um exploring or characterizing the true
nature of sentience or intelligence
here's a really early
2006 or so image from a friston paper
this is the winged Snowflake and it's
like systems that persist conditioned on
seeing this snowflake it's not
conditioned upon all possible you know a
room that's that's 100 degrees C
conditioned on seeing this snowflake
there's going to be snowflakes that
through wind blowing them around
so just doing no better than you'd
expect from the wind currents or doing
better than the wind currents
they preclude phase transition of
melting once they cease to exist then
they cease to exist so that's like one
and then
um because you asked about how to
describe these simple systems too
and then this path in a girl's paper
made the clearest taxonomy of systems
from inert systems where like some of
these nodes are trivial
so some of these nodes can be trivial or
null in effect
and that's what gives the ability to
Model A really simple system
but then the composability of the
Bayesian graph
um format
allows the the construction of also more
sophisticated cognitive models
all keeping the markup blanket

condition because you could have
internal States could just be like a
pass-through or you could like imagine
they functionally didn't exist
or it could just it could be this or it
could be that
um Andrew
I just wanted to add to that that um I
mean in the image after itself later on
I guess section 3.6 page 56
um they attempt to kind of provide their
own answer to the distinction between
like a sentient or perhaps non-sentient
um organism and they just say this is
one answer so not you know an exhaustive
answer to the question but
um like having the capacity for like for
an organism to have the capacity to
model alternative Futures and to have
that level of temporal depth
um that that's one answer that makes
sense to me in thinking about like the
drop of oil on the water like Perhaps it
is you know it's reacting to something
like sensory input and acting but
nonetheless like I can't imagine that
the the oil is you know anticipating
future
um you know sensory information and and
coming up with its own representation of
the proper action to take in the future
and what it should be moving towards so
that's kind of how I've viewed you know
a way of deriving an answer for the
sentience versus non-sentians question
yeah
cool thought and yeah
like as an inner object or even a simple

active object may not provide any
evidence like you don't need to appeal
to any counterfactual generation in
order to explain
even potentially life-like behavior and
certain you know basal cognitive
behavior may not need explicit
entertaining of alternatives
uh wanchon
yes um I have the question about the
figures about the direction of the
arrows can you go back to the figures
3.1 right
so
um if you look here there are um on the
upper right there is uh two uh Direction
figure between active States and
internal States
um
what I'm thinking is that the internal
state
will affect the active stage right so
what action you take depends on your
internal state
but how does active State actually
change the internal stance
directly yeah yeah totally amazing the
internal States will be affected by your
sensory uh sensory
um
stays right
this isn't what are the sensors this
isn't what does happen in a given model
this is what can happen in one
arrangement of an abstract-based graph
so don't connect it to the actual
structure of causality even in the world
this is just saying this is like a

template sketch
of a cybernetic agent with and what's
mainly ruled out
are
the telepathy and telekinesis
and the other one that you can think of
as ruled out instead of asking why this
is two-edged in principle what's ruled
out backwards causation of action we
wanted the entirely in control of action
and backwards causation of sense States
we don't want to have our thumb on the
scale
so there's just many many ways to think
about it and in any actual modeling
situation this level of abstraction is
not engaged with
it becomes a lot clearer when you're
talking about specific functions
describing different states
in a specific system of interest uh
young Hood
oh yeah it's a totally different
question but like considering the
reinforcement learning sometimes when
you sample a lot of sequences and then
like like
there are some models like Tomlin
eichenbaum machine or success
representation which actually updates
the state to state transition based on
the experience that you had
um it it even though you don't really
perceive additional information you can
update the transition metrics based on
your memory so you replay and Playback
the memory and it actually updates your
um internal States so

considering that fact
um where should the where should we put
the memory Replay in that figure
is it like do we have to say that the
memory replay is some kind of um action
or is there another alternative solution
great question
I'll just try to um set it up indirectly
because I think it's a it's a it's a
real
um direction is there's going to be kind
of two
ways Main
families of ways of representing the
cybernetic agent that we're going to see
one of them is this Markov blanket type
so here it's like a very physics base
that's why we're coming from the high
road with the free energy principle
because it starts from this kind of like
flow of physics space
the other representation which we saw in
chapter two and gets used in chapter
four and so on is this
um Bayesian generative model but like
you don't see the
um internal external
and blanket States
though here's different notation but
they're both
um ostensibly about the same active
inference agents right where one can
view the observations here
as the sensory States
and then one can understand that the
actions taken have causal impact on the
world but that's reflected by the B
Matrix and learning on the B Matrix

learning how actions are associated with
changes in the world
so
when it's coming more from like a high
road perspective that's more in the
physics
and that's where you see more of the
physics type formalisms
and then in more of the um uh
decision making and Signal processing
area
from the low road
that's where we see more of these like
just
um
not physics based in principle Bayesian
methods
there's probably a lot more to say about
that and other way to say it but um like
this representation is about a
cybernetic agent
also this representation is about a
cybernetic agents
and they meet inactive inference where
those models have some relationship
okay
figure three points or does anyone else
want to make a comment or question
I wanted to uh just follow up on that
note we look at equation 3.2 a very
similar to equation 2.5 except then 3.2
operational free energies contingent on
a model
so I'm trying to relate that with the
previous comment about the cybernetic
unit in the Markov blanket is that
related to the model
the contingent model

yeah exactly so
free energy like expected free energy
variational free energy
it's a quantity about a model like sum
of squares
is a quantity about a given linear
regression that you might do as like an
error term
if you're fitting a regression
variational free energy is like an error
term that is about a specific model I
mean if you put another node you made a
different model
the free energy would be a different
number
um if you didn't have a model
a map then you can't calculate a free
energy for a model that doesn't exist
so it's a descriptor of a map so
everything is really conditioned upon
the parameters and the values that are
in that specific model
that is
yeah Darius
yeah um just maybe a point of
clarification for myself
I presume it can't be entirely
internalist
um because at least just like looking at
the KL Divergence in the equation
the difference between the approximate
posterior and the true posterior
requires some commitment to their being
an external state that you're trying to
predict
um
so where
is

is that just you've got the variational
model uh so you've got the variational
for energy which is tied to your model
but ultimately it's also on you know uh
subserved by the fact that you are an
agent in the world and that you're
making Bayesian inferences about the
world and it can be closer to the real
world state or not I just don't know if
I've got that the right way around
yes that's true and the kale Divergence
gives us this
optimizable incremental way
to update our beliefs cue
to bring them to
um where where they aren't moving
anymore just as a first approximation
where we we can't believe any better
about perception or action or cognition
with respect to
as you pointed out the the the the
underlying true distribution
but this can be understood in practice
as just not wanting to update your
belief
and
achieving preferred States
which is the maximum evidence model
for
the incoming data
if you literally knew where what photons
were going to come at you next
and that is um
minimized surprise
so this is kind of just like this three
parts
are they're they're being equated here
but they're different framings this is

saying this is the most performant model
best model evidence
this is the least surprising model
if you knew how much to be surprised you
would have the best performing model
this is a well understood
relationship
this is saying because of
how free energy is defined which isn't
invented by active inference lookup
energy-based method or evidence lower
bound
um
this quantity
is tractably optimizable
and if we can reduce the Divergence
um where there's the KL diversions
shrinking will be doing strictly better
that doesn't mean good enough
necessarily but strictly better towards
this
situation
yes
um I have the question about this model
concept does the model refers to others
Dynamic equations in figure 3.1
what what do you mean by model here
here
yeah yeah the model we have about how y
are generated
so so it could renew everything in in
figure 3.1 including like the the
because Y is generated based on your
internal State based on external state
right also it's also affected by the
active State yeah
um
yeah

if you had the total understanding of
what photons would come next
including how the world changes and how
your action influences it
that's the this is like an agent level
loss function
for understanding how y are generated
but you could use energy-based methods
on any Bayesian distribution
but this is just introducing an
important relationship amongst
surprise minimization equals model
evidence maximization
that's the first part and variational
free energy
is a well-used method
to attractively optimize this
um
interface
right so so
3.2 QX is a decision variable so to
minimize you mean or to maximize you
want you want to minimize it right so is
the total beliefs those are the beliefs
that we control but
take it again from from just this level
of the high road
so not explicit counterfactuals yet just
flow across these four states
so the the generation of Y actually are
connected in all those four equations or
four states right how those four states
are changed uh
how this works this changes so M refers
to all the um
all the this process how this process
are described yeah M could be a linear
regression and Y would be the data and

then we'd say
the surprise is minimized the evidence
is maximized and the variational free
energy is minimized
when we have the model and the data and
the models as good as it can be for the
data
so it's the exact same thing like the
two pieces that come into play here are
the model and the data
right so so the decision is your QX all
right so it's a function it's not just
the decision though
QX refers to all of the optimizable
beliefs not just about action
all right so I mean I mean it's it's a
that is a it's a belief that you you may
choose to minimize your total surprise
right or to minimize your variation of
free energy
so among all
you know QX you you you choose one to
minimize this expression on the right
side of 3.2
is that right
so Q x what is X here x does it refer to
the um internal state of their mind
X is the external States
those are the hidden causes about how y
are generated
figure
um 3.1 X external States how sensory
causes are generated no thumb on the
scale
only the external World causes sensory
States
um degenerative model of sensory States
being produced as they are maximize your

evidence about incoming data minimize
your surprise by incoming data minimize
your variational free energy about
incoming data all of those are causal
modeling of external States
analogously to how internal states are
like
doing that for action
because which one you call internal and
external it it you could flip into the
other one
at this level of generality not when you
actually build a model
but this is there's still nothing that's
really fundamentally different between
internal um and external
other than which side the modeler chose
to do internal
Darius
this will be my final question I promise
um I just wanted to ask about the kind
of architecture of the generative model
so the agent is constituted by the
internal States and the Markov blanket
and the agent this was in Ben White's
talk as well there's this point of
potential conflict or confusion that the
agent has a generative model or the
agent is a generative model and that
General generative model has priors
based on its optimism bias of what it
what of the preferred states to
attractive States it wants to be in
um
is it that
can we consider the generative model
therefore
the internal states which are encoding

beliefs about the world and the Markov
Blanca
and in that case how how could you say
if it is that why would you say that the
agent has a genitive model and if you do
hold that if you hold that claim what's
extra about the agent beyond the
genotype model
yeah those are great questions well
what's extra about a person beyond their
health record which you do a linear
regression on I mean the territory
itself so that's the instrumentalist
answer is this is all just about the
model we're just talking about base
graphs and statistics so we're obviously
talking about maps not territories so
that's again trying to take on minimal
commitments but the question about like
does it have a generative model or is it
a generative model well if you're
talking about a software agent you
designed
they might be the same thing you might
say it literally does active inference
because I I actually have it
implementing that here's the script
but if it isn't literally slash actually
an active inference simulation
it seems like you're going to be in the
situation described in figure 9.1
so you're here you're building this is
the figure 4.3 generative model for the
subject of a behavioral experiment
that's why it's subjective it's their
perspective as the Mouse
um in the in the Maze or whatever but
it's also like they are subject to the

behavioral experiment and then here's
the Outer Loop
so would you say that the mouse is the
cognitive model that the researcher made
probably not
so in practice a lot of this like
metaphysics doesn't come into play when
you're talking about a specific
generative model it's a lot clearer
that it's just like a very
um
hopefully well well uh designed
but just a reduced model of the system
of Interest
that chapter nine is about like this
kind of empirical modeling
um
ninky
oh I cannot hear you actually
can you hear me now yeah yeah go for it
okay great
um no I have a question that's very much
related to that I think there are lots
of sort of General comments about the
systems to which we can apply active
inference
um and what sort of the specific form of
criteria are to describe a system as
adaptive
um and I'm also very much interested in
these philosophical perspectives and
seeing sort of how
um in which sense you can you could view
a whole organism
um as
um as sort of following the principles
that specif specified by active
inference or in which sense you could

sort of use active inference to describe
maybe
um
systems at our lower lowering within
that hierarchy that are maybe like
um smaller regulatory systems in the
body
um so I was just wondering if there's
any
um yeah sort of research on uh sort of
the general applicability of active
inference as sort of a way to analyze
adaptive systems
um and I think a lot comes down to the
mathematics which I don't fully
understand yet but
or awesome question that's very cool One
Direction of research that comes to mind
is uh like here's a 2019 paper so
there's been a lot since then and a lot
more integration with active inference
too but um
some of these representations
of um multi-scale
systems
nested systems so being able to
um in like chapter five in the textbook
there's going to be like a motor circuit
that's
influenced by a higher order
decision-making circuit
so then you you maybe you may be able to
to sketch out some of those lower level
units
okay
which paper was this again I'll put it
into the um
I'll put it into the chat

thank you all right well
cool
um about chapter three I guess if uh
next week or in in yeah next week at the
other time we'll come back and if people
can have added or voted on
different questions and we'll talk more
about the questions next week so
thank you
see you all
thank you Daniel thank you